

Lab 9 Handling Text files Part 2

Question 1

```
1/1 + [ ] [ ]
Tilix: pccc@cis106: ~/lab8
1: pccc@cis106: ~/lab8
pccc@cis106:~/lab8$ ls -Am ~ >> file_report.txt
pccc@cis106:~/lab8$ cat file_report.txt
All files in my home directory
.bash_aliases, .bash_history, .bash_history-04214.tmp, .bash_history-06061.tmp,
.bash_logout, .bashrc, .cache, cis106, .config, Desktop, Documents, .dotnet,
Downloads, .face, .face.icon, fonts, for_delete, games, .gitconfig,
git config global user.txt, .gnupg, inClassActivity, lab8, .local, .mozilla,
Music, participation_class, Pictures, .pki, porfolio, practiceonly,
.profile, Public, question2, scripts, .ssh, .sudo_as_admin_successful,
Templates, UsersExport.csv, UsersExport.tsv,
.vboxclient-clipboard-tty2-control.pid, .vboxclient-clipboard-tty2-service.pid,
.vboxclient-draganddrop-tty2-control.pid,
.vboxclient-hostversion-tty2-control.pid, .vboxclient-seamless-tty2-control.pid,
.vboxclient-vmvga-session-tty2-control.pid,
.vboxclient-vmvga-session-tty2-service.pid, Videos, .vscode, website,
website_ch1_lab7, .wget-hsts, wildcard_extra_practice, wildcardpractice
pccc@cis106:~/lab8$

2: pccc@cis106: ~
-A, --almost-all
    print the allocated size of each file, in blocks
-X
    sort alphabetically by entry extension
Written by Richard M. Stallman and David MacKenzie.
or available locally via: info '(coreutils) ls invocation'
pccc@cis106:~$ man ls | "comma"
bash: comma: command not found
pccc@cis106:~$ man ls | grep "comma"
across, horizontal (-x), commas (-m), long (-l), single-column (-1), verbose (-l), vertical (-C)
-H, --dereference-command-line
    follow symbolic links listed on the command line
--dereference-command-line-symlink-to-dir
    follow each command line symbolic link that points to a directory
-m
    fill width with a comma separated list of entries
nected to a terminal. The LS_COLORS environment variable can change the settings. Use the dircolors(1) command to set it.
2
if serious trouble (e.g., cannot access command-line argument).
pccc@cis106:~$
```

Question 2

```
1/1 + [ ] [ ]
Tilix: pccc@cis106: ~/lab8
1: pccc@cis106: ~/lab8
21
pccc@cis106:~/lab8$ ls -lA ~ | grep "^-" | wc -l >> file_report.txt
wc: report.txt: No such file or directory
pccc@cis106:~/lab8$ ls -lA ~ | grep "^-" | wc -l >> file_report.txt
pccc@cis106:~/lab8$ ls -lA ~ | grep "^d" | wc -l >> file_report.txt
pccc@cis106:~/lab8$ tail file_report.txt
.vboxclient-draganddrop-tty2-control.pid,
.vboxclient-hostversion-tty2-control.pid, .vboxclient-seamless-tty2-control.pid,
.vboxclient-vmvga-session-tty2-control.pid,
.vboxclient-vmvga-session-tty2-service.pid, Videos, .vscode, website,
website_ch1_lab7, .wget-hsts, wildcard_extra_practice, wildcardpractice
Total number of files in my home directory
58
Total number of directories in my home directory
21
35
pccc@cis106:~/lab8$

2: pccc@cis106: ~
drwxrwxr-x 2 pccc pccc 4096 Apr 3 11:45 participation_class
drwxr-xr-x 4 pccc pccc 4096 Mar 19 17:52 Pictures
drwx----- 3 pccc pccc 4096 Feb 20 23:29 .pki
drwxrwxr-x 3 pccc pccc 4096 Mar 17 20:52 porfolio
drwxrwxr-x 2 pccc pccc 4096 Apr 3 11:46 practiceonly
drwxr-xr-x 2 pccc pccc 4096 Feb 20 22:14 Public
drwxrwxr-x 4 pccc pccc 4096 Apr 8 22:43 question2
drwxrwxr-x 2 pccc pccc 4096 Mar 14 20:47 scripts
drwx----- 2 pccc pccc 4096 Feb 20 23:10 .ssh
drwxr-xr-x 2 pccc pccc 4096 Feb 20 22:14 Templates
drwxr-xr-x 3 pccc pccc 4096 Mar 12 18:27 Videos
drwxrwxr-x 4 pccc pccc 4096 Feb 20 23:30 .vscode
drwxrwxr-x 3 pccc pccc 4096 Mar 22 13:20 website
drwxrwxr-x 8 pccc pccc 4096 Apr 9 20:57 website_ch1_lab7
drwxrwxr-x 7 pccc pccc 4096 Apr 6 21:32 wildcard_extra_practice
drwxrwxr-x 2 pccc pccc 4096 Apr 8 19:40 wildcardpractice
pccc@cis106:~$
```

Question 3

```
1: pccc@cis106: ~/lab8
dnsmasq /usr/sbin/nologin
avahi /usr/sbin/nologin
speech-dispatcher /bin/false
usbmux /usr/sbin/nologin
cups-pk-helper /usr/sbin/nologin
fwupd-refresh /usr/sbin/nologin
geoclue /usr/sbin/nologin
gnome-remote-desktop /usr/sbin/nologin
saned /usr/sbin/nologin
polkitd /usr/sbin/nologin
rtkit /usr/sbin/nologin
colord /usr/sbin/nologin
Debian-gdm /bin/false
pccc /bin/bash
_flatpak /usr/sbin/nologin
vboxadd /bin/false
pccc@cis106:~/lab8$ awk -F":" '{print $1,$NF}' /etc/passwd
```

```
2: pccc@cis106: ~
pccc@cis106:~$
```

```
3: pccc@cis106: ~
OPTIONS
-F value      sets the field separator, FS, to value.
-f file       Program text is read from file instead of from the command
               line. Multiple -f options are allowed.
-v var=value  assigns value to program variable var.
--            indicates the unambiguous end of options.

The above options will be available with any POSIX compatible implementation
of AWK. Implementation specific options are prefaced with -W. mawk provides
these:

-W dump
Manual page awk(1) line 28 (press h for help or q to quit)
```

```
colord /usr/sbin/nologin /var/lib/colord
Debian-gdm /bin/false /var/lib/gdm3
pccc /bin/bash /home/pccc
_flatpak /usr/sbin/nologin /nonexistent
vboxadd /bin/false /var/run/vboxadd
pccc@cis106:~/lab8$ awk -F":" '{print $1,$NF,$6}' /etc/passwd
```

```
The usercolordlogs in with:./usr/sbin/nologin and its home dir is:/var/lib/colord
The userDebian-gdmlogs in with:./bin/false and its home dir is:/var/lib/gdm3
The userpccclogs in with:./bin/bash and its home dir is:/home/pccc
The user_flatpaklogs in with:./usr/sbin/nologin and its home dir is:/nonexistent
The uservboxaddlogs in with:./bin/false and its home dir is:/var/run/vboxadd
```

```
pccc@cis106:~/lab8$
pccc@cis106:~/lab8$ awk -F":" '{print "The user" $1"logs in with:,"$NF,"and its home dir is:" $6}' /etc/passwd
```

```
2: pccc@cis106: ~
pccc=/bin/bash
_flatpak=/usr/sbin/nologin
vboxadd=/bin/false
pccc@cis106:~/lab8$ awk -F ":" '{print $1 "=" $NF}' /etc/passwd |grep "/bin/bash"
root=/bin/bash
pccc=/bin/bash
pccc@cis106:~/lab8$ awk -F ":" '{print $1 "=" $NF}' /etc/passwd |grep "/bin/bash" >> report_users.txt
pccc@cis106:~/lab8$
```

```
1: pccc@cis106: ~/lab8
The userspeech-dispatcherlogs in with:./bin/false and its home dir is:/run/speech-dispatcher
The userusbmuxlogs in with:./usr/sbin/nologin and its home dir is:/var/lib/usbmux
The usercups-pk-helperlogs in with:./usr/sbin/nologin and its home dir is:/nonexistent
The userfwupd-refreshlogs in with:./usr/sbin/nologin and its home dir is:/var/lib/fwupd
The usergeocluelogs in with:./usr/sbin/nologin and its home dir is:/var/lib/geoclue
The usergnome-remote-desktoplogs in with:./usr/sbin/nologin and its home dir is:/var/lib/gnome-remote-desk
The usersanedlogs in with:./usr/sbin/nologin and its home dir is:/var/lib/saned
The userpolkitdlogs in with:./usr/sbin/nologin and its home dir is:/
The userrtkitlogs in with:./usr/sbin/nologin and its home dir is:/proc
The usercolordlogs in with:./usr/sbin/nologin and its home dir is:/var/lib/colord
The userDebian-gdmlogs in with:./bin/false and its home dir is:/var/lib/gdm3
The userpccclogs in with:./bin/bash and its home dir is:/home/pccc
The user_flatpaklogs in with:./usr/sbin/nologin and its home dir is:/nonexistent
The uservboxaddlogs in with:./bin/false and its home dir is:/var/run/vboxadd
root=/bin/bash
pccc=/bin/bash
pccc@cis106:~/lab8$
```

Question 4

```
1/1 v + [ ] [ ]
Tilix: pccc@cis106: ~/Documents/Csv

1: pccc@cis106: ~/lab8 v
pccc@cis106:~/lab8$ sed 's://,/g' honda1.csv
Car;MPG;Cylinders;Displacement;Horsepower;Weight;Acceleration;Model;Origin
Honda Civic;24.0;4;120.0;97.00;2489.;15.0;74;Japan
Honda Civic CVCC;33.0;4;91.00;53.00;1795.;17.5;75;Japan
Honda Civic CVCC;33.0;4;91.00;53.00;1795.;17.4;76;Japan
Honda Accord CVCC;31.5;4;98.00;68.00;2045.;18.5;77;Japan
Honda Civic CVCC;36.1;4;91.00;60.00;1800.;16.4;78;Japan
Honda Accord LX;29.5;4;98.00;68.00;2135.;16.6;78;Japan
Honda Civic 1500 gl;44.6;4;91.00;67.00;1850.;13.8;80;Japan
Honda Accord;32.4;4;107.0;72.00;2290.;17.0;80;Japan
Honda Civic 1300;35.1;4;81.00;60.00;1760.;16.1;81;Japan
Honda Prelude;33.7;4;107.0;75.00;2210.;14.4;81;Japan
Honda Accord;36.0;4;107.0;75.00;2205.;14.5;82;Japan
Honda Civic;38.0;4;91.00;67.00;1965.;15.0;82;Japan
Honda Civic (auto);32.0;4;91.00;67.00;1965.;15.7;82;Japan
pccc@cis106:~/lab8$ sed 's://,/g' honda1.csv > honda2.csv
pccc@cis106:~/lab8$

2: pccc@cis106: ~/Documents/Csv v
cars.csv cereal.csv contacts.csv country.csv fake_users.csv users_and_ips.csv
pccc@cis106:~/Documents/Csv$ cat ~/lab8/honda1.csv
Car;MPG;Cylinders;Displacement;Horsepower;Weight;Acceleration;Model;Origin
Honda Civic;24.0;4;120.0;97.00;2489.;15.0;74;Japan
Honda Civic CVCC;33.0;4;91.00;53.00;1795.;17.5;75;Japan
Honda Civic CVCC;33.0;4;91.00;53.00;1795.;17.4;76;Japan
Honda Accord CVCC;31.5;4;98.00;68.00;2045.;18.5;77;Japan
Honda Civic CVCC;36.1;4;91.00;60.00;1800.;16.4;78;Japan
Honda Accord LX;29.5;4;98.00;68.00;2135.;16.6;78;Japan
Honda Civic 1500 gl;44.6;4;91.00;67.00;1850.;13.8;80;Japan
Honda Accord;32.4;4;107.0;72.00;2290.;17.0;80;Japan
Honda Civic 1300;35.1;4;81.00;60.00;1760.;16.1;81;Japan
Honda Prelude;33.7;4;107.0;75.00;2210.;14.4;81;Japan
Honda Accord;36.0;4;107.0;75.00;2205.;14.5;82;Japan
Honda Civic;38.0;4;91.00;67.00;1965.;15.0;82;Japan
Honda Civic (auto);32.0;4;91.00;67.00;1965.;15.7;82;Japan
pccc@cis106:~/Documents/Csv$

1/1 v + [ ] [ ]
Tilix: pccc@cis106: ~/lab8

1: pccc@cis106: ~/lab8 v
pccc@cis106:~/lab8$ awk -F ";" 'NR > 1 {print $1,"",$3,"",$5}' honda2.csv | sed 's://,/g' | sed 's/Honda/Honda_/g'
Honda_Civic , 4 , 97.00
Honda_Civic CVCC , 4 , 53.00
Honda_Civic , 4 , 53.00
Honda_Accord CVCC , 4 , 68.00
Honda_Civic CVCC , 4 , 60.00
Honda_Accord LX , 4 , 68.00
Honda_Civic 1500 gl , 4 , 67.00
Honda_Accord , 4 , 72.00
Honda_Civic 1300 , 4 , 60.00
Honda_Prelude , 4 , 75.00
Honda_Accord , 4 , 75.00
Honda_Civic , 4 , 67.00
Honda_Civic (auto) , 4 , 67.00
pccc@cis106:~/lab8$ awk -F ";" 'NR > 1 {print $1,"",$3,"",$5}' honda2.csv | sed 's://,/g' | sed 's/Honda/Honda_/g' >> honda3.csv
pccc@cis106:~/lab8$ batcat honda3.csv
pccc@cis106:~/lab8$

2: pccc@cis106: ~/Documents/Csv v
Unpacking libmbd1s21:amd64 (3.6.5-0.1~deb13u1) ...
Selecting previously unselected package libgit2-1.9:amd64.
Preparing to unpack .../libgit2-1.9_1.9.0+ds-2_amd64.deb ...
Unpacking libgit2-1.9:amd64 (1.9.0+ds-2) ...
Selecting previously unselected package bat.
Preparing to unpack .../bat_0.25.0-2+b2_amd64.deb ...
Unpacking bat (0.25.0-2+b2) ...
Setting up libmbd1s21:amd64 (3.6.5-0.1~deb13u1) ...
Setting up libmbd1s21:amd64 (3.6.5-0.1~deb13u1) ...
Setting up libhttp-parser2.9:amd64 (2.9.4-6+b2) ...
Setting up libgit2-1.9:amd64 (1.9.0+ds-2) ...
Setting up bat (0.25.0-2+b2) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u2) ...
pccc@cis106:~/Documents/Csv$
```

```
Tilix: pccc@cis106: ~/lab8  
t:pccc@cis106: ~/lab8 ▾  
  
File: honda3.csv  
1 model,cylinder,hp  
2 Honda_Civic , 4 , 97.00  
3 Honda_Civic CVCC , 4 , 53.00  
4 Honda_Civic , 4 , 53.00  
5 Honda_Accord CVCC , 4 , 68.00  
6 Honda_Civic CVCC , 4 , 60.00  
7 Honda_Accord LX , 4 , 68.00  
8 Honda_Civic 1500 gl , 4 , 67.00  
9 Honda_Accord , 4 , 72.00  
10 Honda_Civic 1300 , 4 , 60.00  
11 Honda_Prelude , 4 , 75.00  
12 Honda_Accord , 4 , 75.00  
13 Honda_Civic , 4 , 67.00  
:  
  
2: pccc@cis106: ~/Documents/Csv ▾  
Unpacking libmbdts21:amd64 (3.6.5-0.1~deb13u1) ...  
Selecting previously unselected package libgit2-1.9:amd64.  
Preparing to unpack .../libgit2-1.9_1.9.0+ds-2_amd64.deb ...  
Unpacking libgit2-1.9:amd64 (1.9.0+ds-2) ...  
Selecting previously unselected package bat.  
Preparing to unpack .../bat_0.25.0-2+b2_amd64.deb ...  
Unpacking bat (0.25.0-2+b2) ...  
Setting up libmbedx509-7:amd64 (3.6.5-0.1~deb13u1) ...  
Setting up libmbdts21:amd64 (3.6.5-0.1~deb13u1) ...  
Setting up libhttp-parser2.9:amd64 (2.9.4-6+b2) ...  
Setting up libgit2-1.9:amd64 (1.9.0+ds-2) ...  
Setting up bat (0.25.0-2+b2) ...  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for libc-bin (2.41-12+deb13u2) ...  
pccc@cis106:~/Documents/Csv$
```

Challenge Question

```
Applications Apr 17 9:51 PM 24% 55% 0% 0.0 kB 0.0 kB
Tilic: pccc@cis106: ~/lab8/broken
1: pccc@cis106: ~/lab8/broken
pccc@cis106:~/lab8/broken$ ls -lX
styles.css
contact.html
index.html
projects.html
resume.html
favicon.ico
scripts.js
profile.png
pccc@cis106:~/lab8/broken$ firefox index.html
[GFX1-]: RenderCompositorSWGL failed mapping default framebuffer, no dt
pccc@cis106:~/lab8/broken$ grep -E "href | scr" index.html
pccc@cis106:~/lab8/broken$ ls -lX
styles.css
contact.html
index.html
projects.html
resume.html
favicon.ico
scripts.js
profile.png
pccc@cis106:~/lab8/broken$ grep -E "href|scr" index.html
<meta name="description" content="" />
<link rel="icon" type="image/x-icon" href="assets/favicon.ico" />
<link rel="preconnect" href="https://fonts.googleapis.com" />
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
<link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@100;200;300;400;500;600;700;800;900&display=swap" rel="stylesheet" />
<link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.8.1/font/bootstrap-icons.css" rel="stylesheet" />
<link href="css/styles.css" rel="stylesheet" />
<a class="navbar-brand" href="index.html"><span class="fw-bolder text-primary">Start Bootstrap</span></a>
<li class="nav-item"><a class="nav-link" href="index.html">Home</a></li>
<li class="nav-item"><a class="nav-link" href="resume.html">Resume</a></li>
<li class="nav-item"><a class="nav-link" href="projects.html">Projects</a></li>
<li class="nav-item"><a class="nav-link" href="contact.html">Contact</a></li>
<a class="btn btn-primary btn-lg px-5 py-3 me-sm-3 fs-6 fw-bolder" href="resume.html">Resume</a>
<a class="btn btn-outline-dark btn-lg px-5 py-3 fs-6 fw-bolder" href="projects.html">Projects</a>
<a class="text-gradient" href="#"><i class="bi bi-twitter"></i></a>
<a class="text-gradient" href="#"><i class="bi bi-linkedin"></i></a>
<a class="text-gradient" href="#"><i class="bi bi-github"></i></a>
<a class="small" href="#">Privacy</a>
<a class="small" href="#">Terms</a>
<a class="small" href="#">Contact</a>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.2.3/dist/js/bootstrap.bundle.min.js"></script>
<script src="js/scripts.js"></script>
pccc@cis106:~/lab8/broken$
```

```
1: pccc@cis106: ~/lab8/broken
styles.css
contact.html
index.html
projects.html
resume.html
favicon.ico
scripts.js
profile.png
pccc@cis106:~/lab8/broken$ mv styles.css css/
pccc@cis106:~/lab8/broken$ mv scripts.js js/
pccc@cis106:~/lab8/broken$ ls -lX
assets contact.html css favicon.ico index.html js profile.png projects.html res
ume.html
pccc@cis106:~/lab8/broken$ ls -lX
assets
css
js
contact.html
index.html
projects.html
resume.html
favicon.ico
profile.png
pccc@cis106:~/lab8/broken$ mv favicon.ico assets/
pccc@cis106:~/lab8/broken$ ls -lX
assets
css
js
contact.html
index.html
projects.html
resume.html
profile.png
pccc@cis106:~/lab8/broken$ firefox index.html
pccc@cis106:~/lab8/broken$ mv profile.png assets/
pccc@cis106:~/lab8/broken$

2: pccc@cis106: ~/lab8/broken
contact.html
index.html
projects.html
resume.html
profile.png
pccc@cis106:~/lab8/broken$ tree
.
├── assets
│   ├── favicon.ico
│   └── profile.png
├── contact.html
├── css
│   └── styles.css
├── index.html
├── js
│   └── scripts.js
├── profile.png
├── projects.html
└── resume.html

4 directories, 8 files
pccc@cis106:~/lab8/broken$ tree
.
├── assets
│   ├── favicon.ico
│   └── profile.png
├── contact.html
├── css
│   └── styles.css
├── index.html
├── js
│   └── scripts.js
├── projects.html
└── resume.html

4 directories, 8 files
pccc@cis106:~/lab8/broken$

1: pccc@cis106: ~/lab8/broken
contact.html
index.html
projects.html
resume.html
favicon.ico
scripts.js
profile.png
pccc@cis106:~/lab8/broken$ mv styles.css css/
pccc@cis106:~/lab8/broken$ mv scripts.js js/
pccc@cis106:~/lab8/broken$ ls -lX
assets contact.html css favicon.ico index.html js profile.png projects.html res
ume.html
pccc@cis106:~/lab8/broken$ ls -lX
assets
css
js
contact.html
index.html
projects.html
resume.html
favicon.ico
profile.png
pccc@cis106:~/lab8/broken$ mv favicon.ico assets/
pccc@cis106:~/lab8/broken$ ls -lX
assets
css
js
contact.html
index.html
projects.html
resume.html
profile.png
pccc@cis106:~/lab8/broken$ mv profile.png assets/
pccc@cis106:~/lab8/broken$ mv profile.png assets/
pccc@cis106:~/lab8/broken$ firefox index.html
pccc@cis106:~/lab8/broken$ firefox index.html
pccc@cis106:~/lab8/broken$ firefox index.html
pccc@cis106:~/lab8/broken$ firefox index.html

2: pccc@cis106: ~/lab8/broken
contact.html
index.html
projects.html
resume.html
profile.png
pccc@cis106:~/lab8/broken$ tree
.
├── assets
│   ├── favicon.ico
│   └── profile.png
├── contact.html
├── css
│   └── styles.css
├── index.html
├── js
│   └── scripts.js
├── profile.png
├── projects.html
└── resume.html

4 directories, 8 files
pccc@cis106:~/lab8/broken$ tree
.
├── assets
│   ├── favicon.ico
│   └── profile.png
├── contact.html
├── css
│   └── styles.css
├── index.html
├── js
│   └── scripts.js
├── projects.html
└── resume.html

4 directories, 8 files
pccc@cis106:~/lab8/broken$
```



